

14.5cm

STROKAN FC TABLET 250MG

DESCRIPTION: A round, white to off-white in colour, coated tablet, scored on one side with marking 'DUO 5' and 'DUO 6' on the other side.

COMPOSITION: Each tablet contains Ticlopidine HCl 250mg.

PHARMACODYNAMICS: Mechanism of Action: Ticlopidine HCl is an inhibitor of platelet aggregation. When taken orally it causes a time and dose-dependent inhibition of platelet aggregation and release of platelet factors, as well as a prolongation of bleeding time. The drug has no significant in vitro activity. There is no evidence for a circulating active metabolite. The exact mechanism of action is not fully characterised, but does not involve inhibition of the prostacyclin / thromboxane pathways. Platelet cyclic AMP does not seem to play a role in the mechanism of action of Ticlopidine HCl. Ticlopidine HCl interferes with platelet membrane function by inhibiting ADP-induced platelet-fibrinogen binding and subsequent platelet-platelet interactions. Ticlopidine HCl does not act upon cyclo-oxygenase. The effect of Ticlopidine HCl on platelet function is irreversible as shown both by inhibition of fibrinogen binding after washing and by inhibition of platelet aggregation after resuspension of platelets in buffered medium. It is unclear how ticlopidine brings about these changes, a suggested mechanism is that it inhibits ADP-induced exposure of the fibrinogen-binding site of the glycoprotein **IIb-IIIa** complex.

Pharmacodynamics: Inhibition of platelet aggregation is detected within 4 days of administration of a dosage regimen of 250 mg **BD**. Maximum platelet aggregation inhibition (60 to 70%) is achieved 8 to 11 days following dosing with 250 mg **BD**. At the therapeutic dose, ADP-induced platelet aggregation is inhibited by 50 to 70%. Lower total daily doses of 375 mg and 250 mg result in 30 to 60% and 25 to 50% inhibition of platelet aggregation, respectively. Template bleeding time is usually prolonged by two to five-fold of baseline values with the therapeutic dose of Ticlopidine HCl. Upon discontinuation of Ticlopidine HCl dosing, bleeding time and other platelet function tests return to normal within two weeks in the majority of patients. At the recommended therapeutic dose, Ticlopidine HCl has no known significant pharmacological actions in man other than inhibition of platelet function.

PHARMACOKINETICS: After oral administration of the therapeutic dose of Ticlopidine HCl, rapid absorption occurs, with peak plasma levels occurring at approximately 2 hours after dosing. Absorption is at least 80% complete. Administration of Ticlopidine HCl after meals results in an increased (20%) level of ticlopidine in plasma. Ticlopidine hydrochloride displays non-linear pharmacokinetics and clearance decreases markedly on repeated dosing. In older volunteers the apparent half-life of ticlopidine after a single 250 mg dose is about 12.6 hours, with repeat dosing at 250 mg **BD**, the terminal elimination half-life rises to 4 to 5 days and steady state levels of ticlopidine hydrochloride in plasma are obtained after approximately 14 to 21 days. Ticlopidine binds reversibly (88%) to plasma proteins, mainly to serum albumin and lipoproteins. The binding to albumin and lipoproteins is nonsaturable over a wide concentration range. Ticlopidine also binds in a saturable manner to alpha-1 acid glycoprotein. At concentrations attained with the recommended dose, only 15% or less ticlopidine in plasma is bound to this protein. Ticlopidine is metabolised extensively by the liver; no intact ticlopidine is detected in the urine. Following an oral dose of radioactive Ticlopidine HCl administered in solution, 60% of the radioactivity was recovered in the urine and 23% in the faeces. Unmetabolised ticlopidine is a minor component in plasma after a single dose, but at steady state, ticlopidine is the major component. Approximately 40 to 50% of the radioactive metabolites circulating in plasma are covalently bound to plasma proteins. Clearance of ticlopidine decreases with age.

Patients with Liver Impairment: Impaired hepatic function resulted in higher than normal levels of unchanged ticlopidine following single or multiple doses.

Patients with Renal Impairment: Patients with mildly (Cr 50 to 80 mL/min) or moderately (Cr 20 to 50 mL/min) impaired renal function were compared to normal subjects (Cr 80 to 150 mL/min) in a published study of the pharmacokinetic and platelet pharmacodynamic effects of Ticlopidine HCl (250 mg **BD**) for 11 days. Concentrations of unchanged Ticlopidine HCl were measured after a single 250 mg dose and after the final 250 mg dose on Day 11. AUC values of ticlopidine increased by 28 and 60% in mild and moderately impaired patients respectively and plasma clearance decreased by 37 and 52% respectively, but there were no statistically significant differences in ADP-induced platelet aggregation. In this small study (26 patients), bleeding times showed significant prolongation only in the moderately impaired patients.

INDICATIONS: Maintenance of patency of coronary artery by-pass grafts, Maintenance of patency of access sites in patients on chronic haemodialysis, Reducing risk of thrombotic stroke (fatal or nonfatal) in patients who have experienced stroke precursors and in patients who had a completed stroke, Prevention of subacute occlusion following coronary STENT(s) implantation.

RECOMMENDED DOSAGE:
The recommended dose of Ticlopidine HCl is 250 mg twice daily with food. Ticlopidine HCl should be taken with meals to minimise gastrointestinal intolerance.

CONTRAINDICATIONS:
Strokan FC is contraindicated in the following cases:
-Hypersensitivity to the drug or its excipients
-Presence of haematopoietic disorders such as neutropenia, agranulocytosis and thrombocytopenia
-Presence of a haemostatic disorder or of active pathological bleeding (e.g. bleeding peptic ulcer or intracranial bleeding)
-Patients with severe liver impairment or cholestatic jaundice
-Elderly patients (> 65 years) with renal and/or hepatic impairment
-Patients with renal disease on haemodialysis
-Patients with severe heart failure with hepatic congestion.

WARNING & PRECAUTIONS: Use with caution in patients who may be at risk of increased bleeding. Consider discontinuing 10-14 days before elective surgery. Use caution in mixing with other antiplatelet drugs. Use with caution in patients with severe liver disease (experience is limited). May cause life-threatening hematologic reactions, including neutropenia, agranulocytosis, thrombotic thrombocytopenia purpura (TTP), and aplastic anemia. Routine monitoring is required (see monitoring parameters). Monitor for signs and symptoms of neutropenia including WBC count. Discontinue if the absolute neutrophil count falls to <1200/mm or if the platelet count falls to <80,000/mm3.

INTERACTION WITH OTHER MEDICATIONS: Oral anticoagulants and heparins: The tolerance and safety of coadministration of Ticlopidine HCl with heparin, oral anticoagulants or fibrinolytic agents have not been established when used in patients with a high risk of thromboembolic stroke following ischaemic cerebrovascular disturbance. If it is desired to switch a patient from an anticoagulant drug to Ticlopidine HCl, the former drug should be discontinued and its haemostatic effects fully subsided prior to initiation of Ticlopidine HCl therapy. However, the tolerance and safety of coadministration of Ticlopidine HCl with heparin has been studied following the placement of coronary artery stents. Ticlopidine HCl was started immediately after the procedure and heparin was discontinued 12 hours after stent placement.

Aspirin: Aspirin did not modify the ticlopidine-mediated inhibition of ADP induced platelet aggregation, however, ticlopidine potentiates the effect of aspirin on collagen-induced platelet aggregation. The tolerance and safety of antiplatelet doses of aspirin (100-325 mg daily) in combination Ticlopidine HCl 250 mg **BD** has been investigated in patients receiving coronary artery stents for a period of one month. The tolerance and safety of combined therapy at higher doses or periods greater than one month has not been established.

Antiplatelet drugs: Concomitant use with other antiplatelet agents including aspirin carries an increased risk of haemorrhage and require very close monitoring.

Antipyrene: Antipyrene is a model for drugs metabolised by the microsomal hepatic enzyme system (cytochrome p-450 system). Therapeutic doses of ticlopidine caused a 30% increase in the plasma half-life of antipyrene and may cause analogous effects on similarly metabolised drugs. Therefore, dosage of drugs metabolised by hepatic microsomal enzymes with low therapeutic ratios or in patients with hepatic impairment may require adjustment to maintain optimal therapeutic blood levels when starting or stopping concomitant therapy with ticlopidine.

Theophylline: In healthy volunteers stated in published study, concomitant administration of Ticlopidine HCl resulted in a significant increase in the theophylline elimination half-life from 8,6 to 12,2 hr and a comparable reduction in total plasma clearance of theophylline. Theophylline dosage must be adjusted during and after treatment with ticlopidine

Digoxin: Co-administration of Ticlopidine HCl with digoxin resulted in a slight decrease (approximately 15%) in levels of digoxin in plasma. Little or no change in therapeutic efficacy of digoxin would be expected.

Cimetidine: Chronic administration of cimetidine reduced the clearance of a single dose of Ticlopidine HCl by approximately 50%.

Antacids: Administration of Ticlopidine HCl after antacids resulted in a 20% decrease in plasma levels of Ticlopidine HCl.

Phenobarbital: In a small number of healthy volunteers stated in published study, the inhibitory effects of ticlopidine on platelet aggregation were not altered by chronic administration of phenobarbital.

Phenytoin: Cases of elevated phenytoin plasma levels, somnolence and lethargy have been reported during co-administration with Ticlopidine HCl. Caution should be exercised in coadministering this drug with Ticlopidine HCl and it is advised to remeasure phenytoin blood concentrations.NSAIDs: There is an increased risk of haemorrhage with the concurrent use of NSAIDs. If such drugs are necessary clinical and laboratory monitoring (including skin bleeding time) is required.

Plasma Protein Binding: In vitro published studies demonstrate that ticlopidine does not interact with propranolol or phenytoin, basic drugs that are also highly protein bound, with respect to plasma protein binding.

Other Concomitant Therapy: Although specific interaction studies were not performed, in published clinical studies, ticlopidine was used concomitantly with beta blockers, calcium channel blockers and diuretics without evidence of clinically significant adverse interactions.

Carbamazepine: Increased carbamazepine levels and consequent neurological symptoms (dizziness, drowsiness and ataxia) with concomitant administration of ticlopidine has been reported. Patients should be monitored for signs of carbamazepine toxicity when these drugs are taken in combination, Plasma carbamazepine levels should monitored and carbamazepine dosage adjusted when ticlopidine is added or withdrawn.

Cytochrome P450 Effect: CYP3A3/4 enzyme substrate: CYP1A2 (possible, 2C19 9potent), 2D6 (weak) enzyme inhibitor

Increase Effect/Toxicity: Ticlopidine may increase effect/toxicity of aspirin, anticoag lants, theophylline, and NSAIDs. Cimetidine may increase ticlopidine blood levels. Phesytoin blood levels may be increase by ticlopine (case report).

Decrease Effect: Decreased effect of ticlopidine with antacids (decrease absorption). Ticlopine may decrease the effect of diroxin, cyclosporine.

USE IN PREGNANCY & LACTATION: Use in Pregnancy: Ticlopidine HCl is not recommended for use in pregnancy. Published reproductive studies have been performed in mice and rabbits (oral doses up to 200 mg/kg/day) and rats (oral doses up to 400 mg/kg/day) and have revealed no evidence of impaired fertility or teratogenic effects. At the highest doses tested there were increased incidences of resorptions and reduced litter sizes, but this is thought to be due to maternal toxicity rather than a direct effect on the foetus.

Use in Lactation: Ticlopidine HCl is not recommended for use in nursing mothers. Ticlopidine administration to rats during the lactation period prejudiced the survival of pups of mothers receiving the 400 mg/kg/day dose. However, the reproductive performance, postnatal differentiation and functional behavioural tests did not show any changes attributable to ticlopidine. Consistent with the reduced survival of rat pups from mothers receiving ticlopidine during the lactation period was the finding that ticlopidine is excreted into milk. Six hours after a 25 mg/kg oral dose of 14C-ticlopidine the milk concentration to maternal blood plasma concentration ratio was 8,0 and this ratio was maintained for 8 hours, indicating that accumulation occurs in milk.

SIDE EFFECTS: Gastrointestinal: Ticlopidine HCl therapy has been associated with a variety of gastrointestinal complaints including diarrhoea and nausea. The majority of cases are mild, but about 13% of patients discontinued therapy because of these. They usually occur within 3 months of initiation of therapy and typically are resolved within 1 to 2 weeks without discontinuation of therapy. Very rare cases of severe diarrhoea with colitis (including lymphocytic colitis) have been reported. If the effect is severe or persistent, therapy should be discontinued.

Haemorrhage: Ticlopidine HCl has been associated with a number of bleeding complications, mainly bruising or ecchymosis and epistaxis. Haematuria, conjunctival haemorrhage, gastrointestinal bleeding, and peri-operative bleeding and spontaneous post-traumatic bleeding have been reported. Intracerebral bleeding was rare in published clinical trials with Ticlopidine HCl, with an incidence no greater than seen with comparator agents (ticlopidine 0.5%, aspirin 0.6%, placebo 0.75%). The incidence of significant bleeding ranged from 0 to 5.5 percent when Ticlopidine HCl was used for four weeks as adjunctive therapy after coronary artery stenting.

Rash: Ticlopidine HCl has been associated with a maculopapular or urticarial rash (often with pruritus). Rash usually occurs within 3 months of initiation of therapy, with a mean onset time of 11 days. If drug is discontinued, recovery should occur within several days. Many rashes do not recur on drug rechallenge. There have been rare reports of severe rashes including Stevens-Johnson syndrome, erythema multiforme and exfoliative dermatitis.

Less Frequent Adverse Reactions (Probably Related):
Digestive System: hepatitis (cytolytic and cholestatic) during the first months of treatment, GI fullness.
Skin and Appendages: urticaria.
Nervous System: headache.
Body as a Whole: asthenia, pain.
Haemostatic System: epistaxis.
Special Senses: tinnitus.

In addition, rare or very rare: Aplastic anaemia, haemolytic anaemia with reticulocytosis, allergic reactions, anaphylaxis, Quincke edema, arthralgia, hypersensitivity nephropathy, allergic pneumopathy, allergic pneumonitis, systemic lupus (postive ANA), peripheral neuropathy, vasculitis, serum sickness, arthropathy, hepatitis, cholestatic jaundice, angioedema, fever, nephrotic syndrome, myositis, hyponatremia, isolated thrombocytopenia, immune thrombocytopenia, thrombocytopenic thrombotic purpura (TTP), renal failure, hepatic necrosis, sepsis, hepatocellular jaundice, myalgia, cramps, interstitial nephritis, myalgia, cramps and interstitial nephritis.

Altered Laboratory Findings:
Haematological: Neutropenia, thrombocytopenia, agranulocytosis, eosinophilia, pancytopenia, thrombocytosis and bone marrow depression have been reported.
Liver: Ticlopidine HCl therapy has been associated with mild elevations of alkaline phosphatase and transaminases, which generally occur within 1 to 4 months of therapy initiation. No progressive increases were seen in closely monitored clinical trials, but most patients with these abnormalities had therapy discontinued. Occasionally, patients had developed minor deviations in bilirubin.
Cholesterol: Ticlopidine HCl therapy has been associated with increased serum cholesterol and triglycerides. Serum levels of HDL-C, LDL-C, VLDL-C, and triglycerides are increased 8 - 10% within 1 month of therapy and persist at those levels. The ratios of the lipoprotein subfractions are unchanged. The effect is not correlated with age, sex, alcohol use, or diabetes.

SYMPTOMS AND TREATMENT OF OVERDOSE: Following overdosage, gastric lavage and other general supportive measures are recommended. Coagulation parameters should be checked; the patient should be monitored closely for evidence of bleeding (including intracranial and other occult bleeding); full blood count (including platelet count) should be monitored for up to two weeks after the overdose.

STORAGE CONDITIONS: Store below 30° C. Protect from light. Keep out of reach of children. Jauhkan daripada kanak-kanak.

PACK SIZE: 2 x10's, 3x10's tablets in blister pack

SHELF LIFE: Please refer to outer package

ROUTE OF ADMINISTRATION: Oral

PRODUCT REGISTRATION HOLDER & MANUFACTURER:
DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA
1500012531 N.2

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